

ALEXANDER G. LUCACI, PH.D.

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Computational biologist studying microbial evolution in complex environments. My work develops scalable phylogenetic, metagenomic, and multi-omic methods to detect selection, characterize microbiomes, and support genomic surveillance across global urban and public-health systems.

EDUCATION

Ph.D. in Bioinformatics , Temple University	2023
<i>Dissertation: The Role of Complex Evolutionary Dynamics in Molecular Sequence Analysis</i>	
M.S. in Biology , New York University	2018
B.S. in Biochemistry , Stony Brook University	2011

RESEARCH APPOINTMENTS

Weill Cornell Medical College	July 2023 to Present
<i>Postdoctoral Associate</i>	<i>Advisor: Dr. Christopher E. Mason</i>

- Conduct interdisciplinary research at the interface of evolutionary biology, metagenomics, and viromics, with emphasis on the NIH Human Virome Program (HVP) and the MetaSUB Consortium. Apply molecular sequence analysis and evolutionary theory to large-scale genomic and metatranscriptomic datasets to characterize urban microbial and viral ecosystems.
- Develop scalable computational frameworks for selection analysis, urban virome surveillance, and real-time biological threat detection, contributing to global efforts in city-scale microbiome monitoring and predictive evolutionary modeling.

Temple University	Spring 2020 to Spring 2023
<i>Graduate Research Assistant</i>	<i>Advisor: Dr. Sergei L. Kosakovsky Pond</i>

- Designed and implemented advanced computational methods in HyPhy to quantify the impact of multi-nucleotide substitutions (MNS) on molecular adaptation, contributing to the BUSTED+S+MH and FMM frameworks that explicitly model clustered mutational processes and improve the robustness of episodic and gene-wide dN/dS inference.

RESEARCH LEADERSHIP

NIH - Human Virome Program	January 2025 to Present
<i>Co-Chair: Bioinformatics Working Group</i>	

Coordinate bioinformatics strategy for the Human Virome Program initiative, developing standardized pipelines for viromics analysis and scalable methods for viral discovery and host profiling. Drive cross-disciplinary collaboration, multi-omic integration, and reproducible, open viral bioinformatics infrastructure.

The MetaSUB International Consortium	June 2024 to Present
<i>Executive Director / Director of Bioinformatics</i>	

Oversee global bioinformatics workflows for MetaSUB urban microbiome studies across 100+ cities, developing scalable pipelines for taxonomic, functional, AMR, and viral analyses. Coordinate international data harmonization, mentor trainees, and advance computational methods for environmental and public health microbiology.

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2. Gao Z*, Wu J*, **Lucaci AG***, Ouyang J, Wang L, Ryon K, Elhaik E, Probst AJ, Rod X, Velavan TN, Chasapi A, Ouzounis CA, Oliveira M, Dias-Neto E, Osuolale OO, Poulsen M, Meleshko D, Bhattacharyya M, Ugalde JA, Sierra MA, Tierney BT, Prithiviraj B, Sharma NK, Munteanu V, Mangul S, Ushio M, abaj PP, Toscan R, Subramanian B, Frolova A, Burkhart J, Deng Y, Udekwu KI, Schriml LM, Hazrin-Chong NH, Suzuki H, Lee PKH, Wang LF, Mason CE, Shi T. Diversity and Distinctive Traits of the Global RNA Virome in Urban Environments. *SSRN*. 2024. doi: 10.2139/ssrn.4871972. **Accepted at Nature Communications**
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4. **Lucaci AG**, Pond SLK. (2026). AOC: A Snakemake workflow for the characterization of natural selection in protein-coding genes. *Journal of Open Source Software*, 11(121), 9872, <https://doi.org/10.21105/joss.09872>
5. Sharma, G., Munteanu, V., Ghiasi, N. M., Mahanta, U., Banerjee, J., Varma, S., Foschini, L., Ellrott, K., Mutlu, O., Ciorb, D., Ophoff, R. A., Bostan, V., Moore, J. H., Sousoni, D., Krishnan, A., **Lucaci, AG**, Tull, A., Mason, C. E., Dimian, M., Mangul, S. (2026). Towards a decentralized future for open-science databases. *Nature Genetics*, 15. <https://doi.org/10.1038/s41588-026-02606-x>
6. Mak L, Tierney B, Wei W, Ronkowski C, Toscan RB, Turhan B, Toomey M, Andrade-Martinez JS, Fu C, **Lucaci AG**, Solano AHB, Setubal JC, Henriksen JR, Zimmerman S, Kopbayeva M, Noyvert A, Iwan Z, Kar S, Nakazawa N, Meleshko D, Horyslavets D, Kantsypa V, Frolova A, Kahles A, Danko D, Elhaik E, Labaj P, Mangul S; International MetaSUB Consortium; Mason CE, Hajirasouliha I. CAMP: a modular metagenomics analysis system for integrated multistep data exploration. *NAR Genom Bioinform*. 2026 Jan 16;8(1):lqaf172. doi: 10.1093/nargab/lqaf172. PMID: 41551931; PMCID: PMC12809600.
7. Zhu K, Amirali A, Auch B, Babler KM, Biswas P, Bowie K, Choudhary S, Currall BB, Grills GS, Healy HG, Liachko I, **Lucaci AG**, Mason CE, Sharkey M, Shigeno Risse-Adams O, Shukla BS, Sisson Z, Stevenson M, Williams SL, Zulli A, Peccia J, Solo-Gabriele HM. Proof-of-concept of host attribution of antimicrobial resistance genes using wastewater Hi-C metagenome sequencing. *J Water Health*. 2026 Feb;24(2):148-159. doi: 10.2166/wh.2025.162. Epub 2025 Dec 29. PMID: 41764388.
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11. Selberg, A., Clark, N. L., Sackton, T. B., Muse, S. V., **Lucaci, AG**, Weaver, S., Nekrutenko, A., Chikina, M., & Pond, S. L. K. (2025). Minus the Error: Testing for Positive Selection in the Presence of Residual Alignment Errors. *eLife*, 14. <https://doi.org/10.7554/eLife.106921.1>
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16. Overbey EG, Kim J, Tierney BT, Park J, Huerbi N, **Lucaci AG**, Garcia Medina S, Damle N, Najjar D, Grigorev K, Afshin EE, Ryon KA, Sienkiewicz K, Patras L, Klotz R, Ortiz V, MacKay M, Schweickart A, Chin CR, Sierra MA, Valenzuela MF, Dantas E, Nelson TM, Cekanaviciute E, Deards G, Foox J, Narayanan SA, Schmidt CM, Schmidt MA, Schmidt JC, Mullane S, Tigchelaar SS, Levitte S, Westover C, Bhattacharya C, Lucotti S, Wain Hirschberg J, Proszynski J, Burke M, Kleinman AS, Butler DJ, Loy C, Mzava O, Lenz J, Paul D, Mozsary C, Sanders LM, Taylor LE, Patel CO, Khan SA, Suhail Mohamad M, Byhaqui SGA, Aslam B, Gajadhar AS, Williamson L, Tandel P, Yang Q, Chu J, Benz RW, Siddiqui A, Hornburg D, Bleas K, Moreno J, Boddicker A, Zhao J, Lajoie B, Scott RT, Gilbert RR, Lai Polo SH, Altomare A, Kruglyak S, Levy S, Ariyapala I, Beer J, Zhang B, Hudson BM, Rininger A, Church SE, Beheshti A, Church GM, Smith SM, Crucian BE, Zwart SR, Matei I, Lyden DC, Garrett-Bakelman F, Krumsiek J, Chen Q, Miller D, Shuga J, Williams S, Nemec C, Trudel G, Pelchat M, Laneuville O, De Vlaminc I, Gross S, Bolton KL, Bailey SM, Granstein R, Furman D, Melnick AM, Costes SV, Shirah B, Yu M, Menon AS, Mateus J, Meydan C, Mason CE. The Space Omics and Medical Atlas (SOMA) and international astronaut biobank. *Nature.* 2024 Aug;632(8027):1145-1154. doi: 10.1038/s41586-024-07639-y. Epub 2024 Jun 11. PMID: 38862028; PMCID: PMC11357981.
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- HspB5 and HspB8: Shaping these disease factors during vertebrate evolution. *Cell Stress Chaperones*. 2022 Jul;27(4):309-323. doi: 10.1007/s12192-022-01268-y. Epub 2022 Jun 9. PMID: 35678958; PMCID: PMC9346038.
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BOOK CHAPTERS

Fundamentals of Space Medicine and Clinical Technology: Viral evolution in the cosmos
Synthesizes evolutionary dynamics of viruses across extreme and extraterrestrial environments. Highlights selection pressures and adaptation in constrained ecosystems using comparative genomics frameworks. Connects viromics and astrobiology to predictive models of biological discovery and planetary health. Positions viral evolution as a framework for understanding biological adaptation in novel and engineered environments.

- Trovao NS, **Lucaci AG**, Pradeep N, Sethi M, Bono LM, Singh RS, Clark KB, Zaksas V, Burke M, Caicedo A, Castaeda V, Zambrano K, Karim R, Theriot CA, Otkiran G, Neefs D, Isola G, Tartaglia G, Schotsaert M, Tamim S, Das S, Fasullo M, Ali N, Fargette D, Brereton NJB, Beheshti A, Guarnieri JW, Wurtele ES. *Viral evolution in the cosmos*. In: *Fundamentals of Space Medicine and Clinical Technology*. Academic Press; **2026**. pp. 67–96. doi: 10.1016/B978-0-443-32904-3.00009-1.

SOFTWARE, DATA RESOURCES, AND WORKFLOWS

1. **RASCL: Rapid Assessment of Selection in Clades** *Lead developer / first author*
Computational workflow for rapid molecular selection analysis across clades, supporting scalable and reproducible evolutionary analysis of protein-coding sequences. Designed to enable rapid assessment of selection patterns across viral and comparative genomic datasets.
Associated publication: **Lucaci AG**, Zehr JD, Shank SD, Bouvier D, Ostrovsky A, Mei H, Nekrutenko A, Martin DP, Kosakovsky Pond SL. RASCL: Rapid Assessment of Selection in CLades through molecular sequence analysis. *PLOS One*. 2022 Nov 2;17(11):e0275623. doi: 10.1371/journal.pone.0275623. PMID: 36322581; PMCID: PMC9629619.
2. **AOC: Analysis of Orthologous Collections** *Lead developer / first author*
Application for organizing orthologous protein-coding sequence collections and characterizing natural selection across genes and taxa. Developed to support reproducible comparative molecular evolutionary analyses across large orthologous datasets.
Associated preprint: **Lucaci AG**, Pond SLK. AOC: Analysis of Orthologous Collections – an application for the characterization of natural selection in protein-coding sequences. *ArXiv [Preprint]*. 2024 Jun 13:arXiv:2406.09522v1. PMID: 38947939; PMCID: PMC11213150.
3. **HyPhy-based molecular evolution methods** *Lead method developer / first author*
Designed and implemented computational approaches for evaluating the impact of multinucleotide substitutions and clustered mutational processes on molecular adaptation inference. Contributed to model development and workflow implementation supporting robust episodic and gene-wide selection analyses, including BUSTED+S+MH and FMM-related frameworks.
Associated publication: **Lucaci AG**, Zehr JD, Enard D, Thornton JW, Kosakovsky Pond SL. Evolutionary Shortcuts via Multinucleotide Substitutions and Their Impact on Natural Selection Analyses. *Molecular Biology and Evolution*. 2023 Jul 5;40(7):msad150. doi: 10.1093/molbev/msad150. PMID: 37395787; PMCID: PMC10336034.
4. **NIH Human Virome Program bioinformatics working group** *Co-Chair*
Coordinate development of standardized and reproducible computational workflows for viromics analysis, viral discovery, host profiling, data harmonization, and public data submission across the NIH Human Virome Program. These workflows support large-scale integration of multi-omic datasets and promote interoperable viral bioinformatics infrastructure.
5. **MetaSUB global urban microbiome workflows** *Director of Bioinformatics*
Direct bioinformatics strategy for large-scale urban microbiome and virome studies across 100+ cities through the MetaSUB International Consortium. Develop and coordinate scalable workflows for taxonomic profiling, functional analysis, antimicrobial resistance detection, viral discovery, and cross-site data harmonization.

6. **Global urban RNA virome analysis workflows** *Co-first author / workflow development*
 Developed and coordinated computational analyses for large-scale characterization of RNA viral diversity in urban environments, including viral discovery, comparative virome profiling, and harmonized analysis across international urban metagenomic datasets.
Associated preprint: Gao Z*, Wu J*, **Lucaci AG***, Ouyang J, Wang L, Ryon K, Elhaik E, Probst AJ, Rod X, Velavan TN, Chasapi A, Ouzounis CA, Oliveira M, Dias-Neto E, Osuolale OO, Poulsen M, Meleshko D, Bhattacharyya M, Ugalde JA, Sierra MA, Tierney BT, Prithiviraj B, Sharma NK, Munteanu V, Mangul S, Ushio M, abaj PP, Toscan R, Subramanian B, Frolova A, Burkhart J, Deng Y, Udekwu KI, Schriml LM, Hazrin-Chong NH, Suzuki H, Lee PKH, Wang LF, Mason CE, Shi T. Diversity and Distinctive Traits of the Global RNA Virome in Urban Environments. *SSRN*. 2024. doi: 10.2139/ssrn.4871972.
7. **Urban virome surveillance and biological threat detection** *Workflow Architect*
 Develop scalable computational frameworks for analyzing metagenomic and metatranscriptomic datasets from urban environments, with applications to city-scale microbial and viral surveillance, environmental genomics, public health monitoring, and real-time biological threat detection.
8. **Reproducible training workflows for global bioinformatics education** *Instructor*
 Developed hands-on instructional workflows for molecular adaptation, metagenomics, sequence analysis, and reproducible computational biology through international workshops and training programs. Materials emphasize practical analysis of real biological datasets using accessible, reproducible computational methods.
Associated preprint: Oleksyk TK, Yakymenko D, Boek S, Munteanu V, Pilch W, Comarova Z, et al. Leveraging a hybrid cross-disciplinary training model to accelerate global bioinformatics capacity [Internet]. *bioRxiv*; 2026 [cited 2026 Feb 15]. p. 2026.01.21.700760.
 Available from: <https://www.biorxiv.org/content/10.64898/2026.01.21.700760v1>

GRANTS, FELLOWSHIPS, AND AWARDS

- **2024** NIH Loan Repayment Program (LRP) Award, *National Cancer Institute*, **\$100,000**
- **2023** U54 (NIH Human Virome Program): Lead Analyst and architect of the Data Analysis & Submission Core for the center grant, enabling large-scale genomic data analysis and public data submission. This role involved designing reproducible computational infrastructure for harmonized virome data processing, metadata integration, and deposition across collaborating sites.
- **2023** Young Investigators Travel Award, *SMBE*, **\$3,500**
- **2023** Opening Influenza Research Fellowship, *Center for Open Science*, **\$2,000**
- **2022** CST Outstanding Research Award, *Temple University*, **\$500**
- **Grant Applications:**
 - NIH VAST Supplement: Proposed development of scalable, AI-enabled frameworks for viral surveillance and genomic data integration supporting the NIH Human Virome Program.
 - Horizon Europe Grant: Developed a multinational proposal to harmonize urban microbiome and virome data across global cities using FAIR, AI-ready data standards and predictive modeling. The proposal emphasized interoperable metadata, reproducible workflows, and comparative analyses across built environments.
 - DOE Genesis Grant: Designed a project to establish standardized, AI-ready multi-omics datasets and workflows for *Pseudomonas putida*, integrating proteomics, metabolomics, and lipidomics with FAIR-compliant metadata. The project framed microbial systems biology as a foundation for reusable, machine-readable datasets that support downstream modeling and discovery.
- **Professional Membership:** Sigma Xi (The Scientific Research Honor Society)
- **Travel Awards:** SMBE (2020, 2023); EMBO (2023); Center for Open Science Metascience (2023)

TEACHING EXPERIENCE

Weill Cornell Medicine: Quantitative Understanding in Biology

Fall 2025

Guest Lecturer

- Delivered an invited module on metagenomics and metatranscriptomics within the course *Quantitative Information in Biological Systems*, integrating real-world urban microbiome case studies and demonstrating computational workflows for high-throughput sequencing data analysis.
- Emphasized quantitative reasoning, reproducible bioinformatics practices, and interpretation of multi-omic datasets for students in quantitative biology and biomedical programs.

International Workshops and Summer Schools

- **Instructor, VEME 2023 & 2024 (International Workshop on Virus Evolution and Molecular Epidemiology)** Designed and delivered a half-day intensive module on molecular adaptation, including hands-on HyPhy-based selection analyses for an international audience of researchers in bioinformatics, virology, and public health.
- **Instructor, EEBG 2024 & 2025 (Eastern European Bioinformatics and Computational Genomics School)** Taught modules on molecular adaptation and metagenomics, combining conceptual lectures with practical training in evolutionary analysis workflows and metagenomic data interpretation.

Temple University Bioinformatics Studio

Spring 2018 to 2023

Co-Founder

- Co-founded and scaled a student-led training initiative focused on accessible, hands-on bioinformatics education for learners ranging from beginners to advanced trainees.
- Developed and delivered workshops covering sequence analysis, metagenomics, and reproducible computational workflows.
- Mentored students and early-career researchers in computational biology best practices, fostering an inclusive peer-learning environment.

Temple University Genomics in Medicine (Graduate TA)

Fall 2019 to 2021

- Supported enrollment of 100–160 students per semester in a cross-listed graduate/undergraduate genomics course.
- Led weekly office hours and review sessions, clarified primary literature, and provided detailed feedback on computational and written assignments.
- Assisted students with project design and interpretation of genomics data, strengthening quantitative and translational understanding.

Temple University Laboratory Instruction (Graduate TA)

2018 to 2019

- **Intro to Organismal Biology (Spring 2019)** Led two laboratory sections (40 students total), instructed experimental techniques, reinforced core biological concepts, and provided individualized feedback on lab performance.
- **General Biology I (Fall 2018)** Directed laboratory sessions for 40 students, demonstrated experimental protocols, and evaluated laboratory reports and practical skills.

New York University Adjunct Professor

2017 to 2018

- **Fundamentals of Bioinformatics (Spring 2018)** Led weekly recitations for 20 students, teaching core bioinformatics theory and software applications while supporting student projects.
- **Molecules of Life (Fall 2017)** Instructed two laboratory sections (40 students total), reinforcing lecture material through hands-on biochemical and molecular biology exercises.
- **Principles of Biology Lab (Spring 2017)** Taught two laboratory sections, guiding students through experimental design, data analysis, and foundational biological concepts.

MENTORING

1. Research Mentor

2025–Present

Mentor trainees across graduate, medical, master's, and high school levels in bioinformatics, metagenomics, viromics, molecular evolution, reproducible computational workflows, and scientific communication. Provide guidance on project design, data analysis, interpretation of high-throughput sequencing datasets, manuscript preparation, and professional development.

- **Oona Shingeo Risse-Adams** – Graduate Student
- **Betty Judah** – Medical Student
- **Nesara Dandiganahalli Lakshmiprasad** – Masters Student
- **Ayeh Khorshidian** – Masters Student
- **Gabriella Smyth** – High School Student

SERVICE

- **Journal Reviewer:** *Astrobiology* (2026); *Bioinformatics Advances* (2026, 2025); *Molecular Biology and Evolution* (2024); *Virus Evolution* (2023); *Genomics* (2021); ad hoc reviewer for journals in evolutionary genomics, virology, and computational biology.
Reviewed manuscripts on molecular evolution, metagenomics, and genomic data analysis
- **Biological Science and Technology (BSAT) Study Group Member,**
Defense Advanced Research Projects Agency (DARPA), 2025–2026
 - Contributed expert input on emerging priorities in biological science and technology, including computational biology, genomics, and biosecurity
 - Advise on future research directions at the intersection of biotechnology, national security, and data science

PRESENTATIONS

Invited Talks

- **2026** American Museum of Natural History (SciCafe). The Subway's invisible microbiome.
- **2025** Clarkson University. Challenges in examining the global urban microbiome.

Conference and Workshop Presentations

- **2026** ICCABS. Diversity and distinctive characteristics of the global RNA virome in urban & peri-urban environments.
- **2025** AGBT. Diversity and distinctive traits of the global RNA virome in urban environments.
- **2025** RdRp Summit. The global RNA virome in urban environments.
- **2025** CAMErA. Challenges in examining the global urban microbiome.
- **2025** NASA GeneLab Microbes AWG. The global RNA virome in urban environments.
- **2024** MetaSUB Consortium. Quantifying natural selection in urban environments.
- **2023** Dynamics and Evolution. Selection analyses identify broad evolutionary features across viral families.
- **2023** Temple University. The role of complex evolutionary dynamics in molecular sequence analysis.
- **2022** Weill Cornell Medical College. Advances in quantifying natural selection in coding sequences.
- **2022** Dynamics and Evolution of Human Viruses. RASCL: Rapid assessment of selection in clades.

- **2021** EMBO. Widespread empirical support for instantaneous multiple-nucleotide changes.
- **2021** Mid-Atlantic Bioinformatics Conference. Rapid assessment of selection in SARS-CoV-2 variants.
- **2020** Cold Spring Harbor Laboratory. Widespread empirical support for instantaneous multiple-nucleotide changes.
- **2020** SMBE. Evaluating the impact of multiple simultaneous mutations on evolutionary rate inference.
- **2019** BGSS Retreat, Temple University. Evaluating the impact of multiple simultaneous mutations on evolutionary rate inference.